

Mitigating Physical Layer Vulnerabilities in 6G Communications using Deep Reinforcement Learning

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Introduction

6G networks promise ultra-high-speed connectivity and massive device density — but the wireless physical layer remains inherently exposed to intelligent adversaries.



6G Promise

Ultra-high speed, ultra-low latency, and massive connectivity for next-generation applications.



Jamming Attacks

Malicious interference that disrupts legitimate communication channels.



Eavesdropping

Passive interception of transmitted signals to extract confidential data.



Static Defense Gap

Traditional fixed security mechanisms cannot adapt to dynamic, evolving attack patterns.

Problem Statement

The Core Problem

Static security approaches are fundamentally incapable of dynamically adapting to the rapidly changing nature of wireless physical layer attacks in 6G environments.

- Fixed countermeasures fail against adaptive adversaries
- No real-time situational awareness
- Inability to optimize under uncertainty

The Critical Need

An **intelligent, adaptive security mechanism** capable of learning from the wireless environment and autonomously adjusting transmission strategies to counter evolving threats.

- ① 6G physical layer security demands real-time, data-driven decision-making beyond the reach of rule-based systems.

Objective of the Study

This research develops a **Deep Reinforcement Learning (DRL)-based adaptive framework** to enhance physical layer security in 6G communication systems under dynamic attack conditions.

1

Improve Secrecy Performance

Maximize the achievable secrecy rate between legitimate transmitter and receiver.

2

Reduce Attack Impact

Minimize the effectiveness of jamming and eavesdropping through adaptive countermeasures.

3

Optimize Transmission

Dynamically select beamforming, power, and artificial noise strategies.

4

Enhance Robustness

Ensure reliable communication under adversarial and time-varying channel conditions.

Related Work

Prior Research Focus Areas

- **RL for Wireless Security** — Q-learning and policy gradient methods applied to anti-jamming.
- **RIS-Assisted Security** — Reconfigurable intelligent surfaces for passive beamforming.
- **Secrecy Rate Optimization** — Game-theoretic and convex optimization approaches.

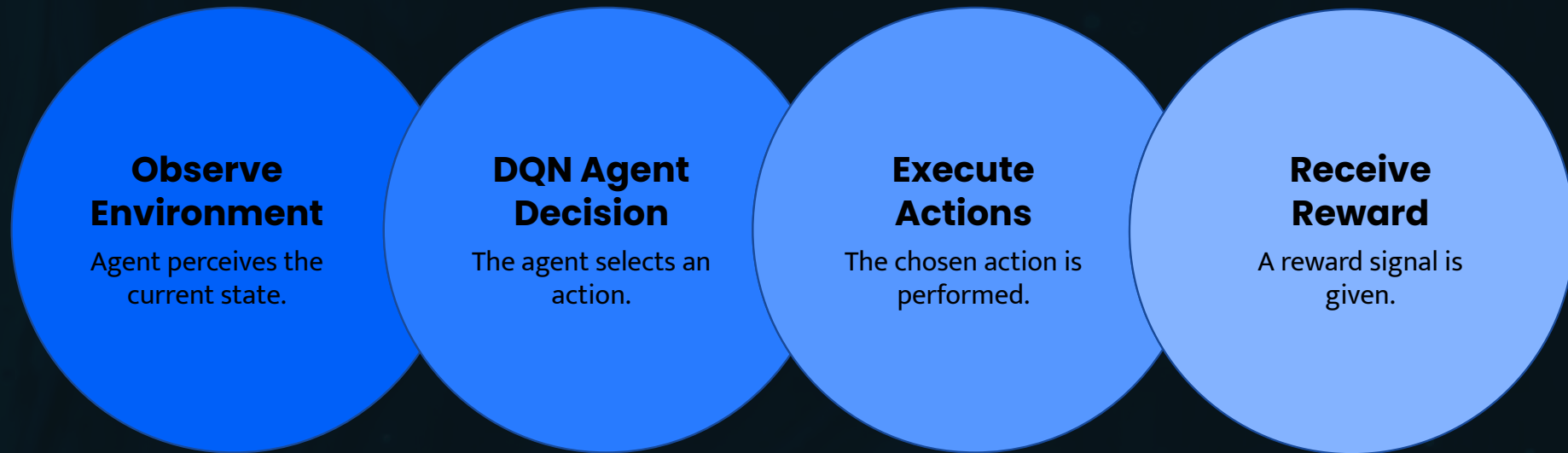
Key Limitation

Most existing studies address **isolated, static scenarios** rather than providing continuous, real-time adaptive protection against evolving multi-threat environments.

⚠ Gap: Lack of end-to-end DRL frameworks that jointly optimize beamforming, power, and artificial noise under dynamic adversarial conditions.

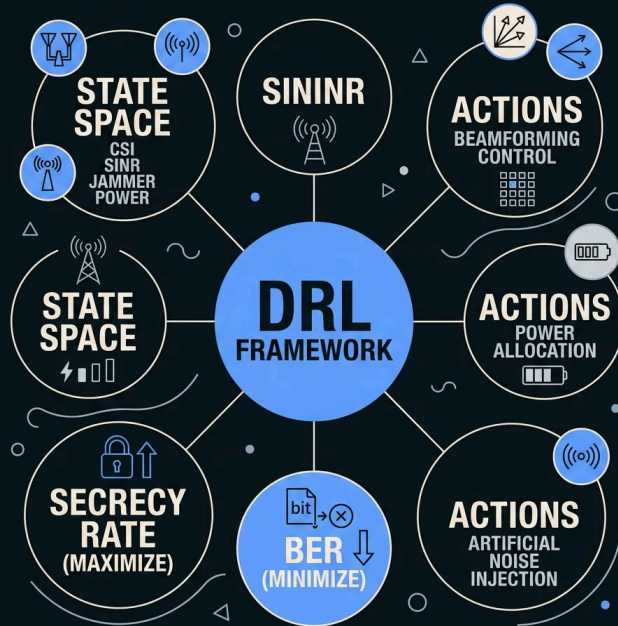
Proposed System Architecture

A **DQN-based adaptive security framework** where the agent continuously observes the wireless environment, selects optimal transmission actions, and learns through reward feedback to counter physical layer attacks.



The closed-loop DRL process enables continuous adaptation: the agent observes channel state, decides on transmission parameters, executes countermeasures, and refines its policy based on secrecy and reliability rewards.

Methodology



DRL Formulation

The problem is modeled as a **Markov Decision Process (MDP)** where the DQN agent maps observed channel and attack states to optimal transmission actions.

State Space

CSI, SINR, Jammer Power — captures full adversarial context.

Action Space

Beamforming vectors, power levels, artificial noise variance.

Reward Function

Weighted combination of Secrecy Rate ($\uparrow$) and BER ($\downarrow$).

Simulation Environment



Experimental Setup

A lightweight Python-based simulation environment was developed to model dynamic 6G wireless conditions under adversarial scenarios.



Dynamic Conditions

Time-varying channels with fading, path loss, and mobility models.



Attack Modeling

Simulated jamming and eavesdropping with adaptive adversary behavior.

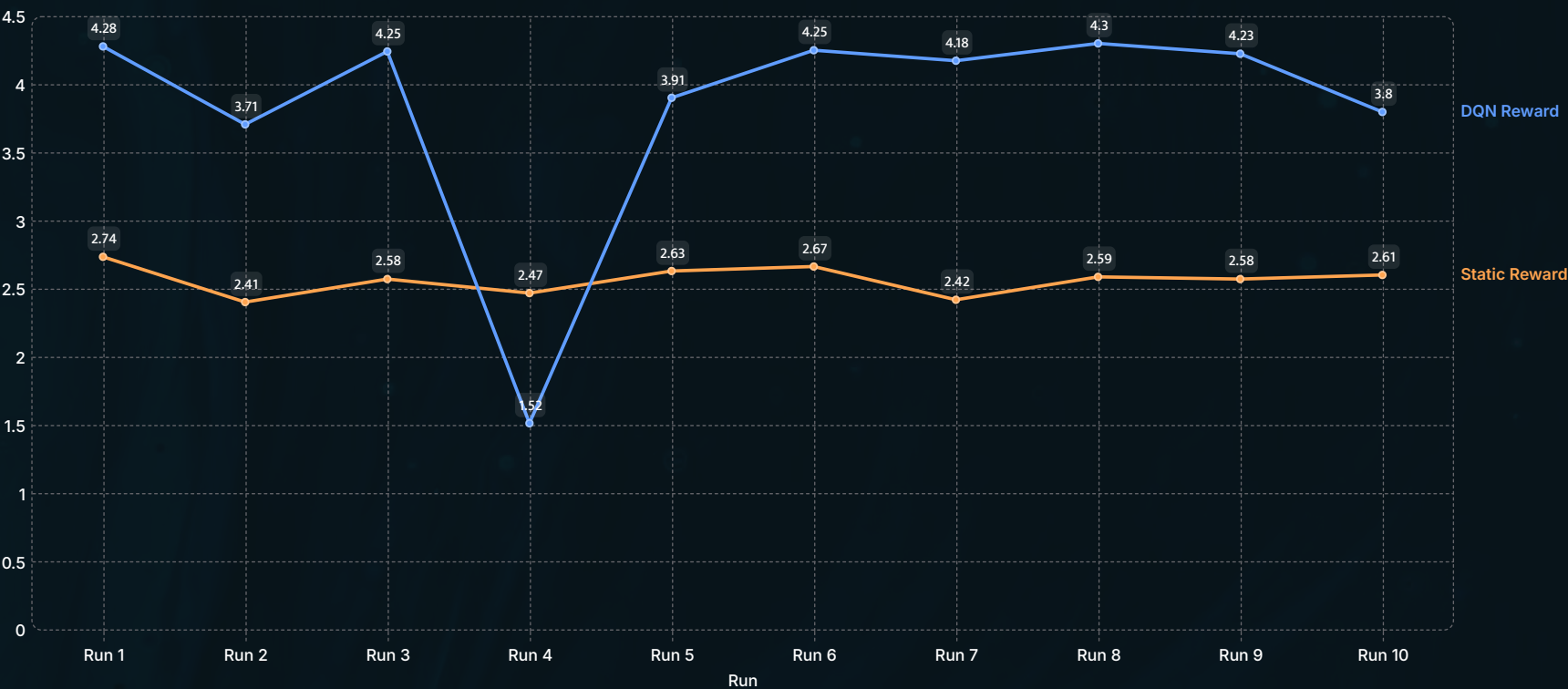


Baseline Comparison

DQN performance benchmarked against a static non-adaptive transmission strategy.

Experimental Results

Comparison of DQN Reward and Static Reward across 10 simulation runs in dynamic adversarial 6G wireless conditions.



Key Findings

3.84

Average DQN Reward

Mean cumulative reward achieved by the proposed DQN agent.

2.57

Average Static Reward

Mean reward under the non-adaptive baseline strategy.

49.28%

Performance Improvement

Relative gain in reward demonstrating the adaptive advantage of DQN.

Conclusion: Across all 10 runs, DQN outperforms the static baseline in most simulation runs, demonstrating strong adaptive performance in adversarial 6G environments.ts.

Conclusion & Future Work

Conclusions



DRL enables **real-time adaptive security** against dynamic physical layer attacks.



The DQN agent effectively **mitigates jamming and eavesdropping** through joint optimization.



Secrecy-oriented performance improved by **~49%** over static baselines.

Future Directions

Advanced Algorithms

Explore PPO and DDPG for continuous action spaces and faster convergence.

Multi-Agent RL

Coordinate multiple DRL agents for distributed network-wide security.

RIS Integration

Incorporate reconfigurable intelligent surfaces for enhanced beamforming control.

Large-Scale 6G

Validate framework scalability in realistic, multi-cell 6G deployment scenarios.

Thank You!

Questions & Discussion

We appreciate your attention and insights.